RemoVia

Setup & TV Pairing Guide

RemoVia is one remote on your phone for every smart TV in the house - Roku, Samsung, Google TV, Android TV and Fire TV. This guide takes you from download to a working remote, one step at a time. Anyone can follow it.

The remote page itself lives at removia.iecincglobal.com. This guide is also on the web at iecincglobal.com/removia-guide.html.

Step 1 - What you need

- A Mac on the same Wi-Fi as your TVs. A small program called the bridge runs on the Mac. It finds your TVs and talks to each one in its own language. The Mac needs Python 3.11 or newer - the bridge sets up the rest by itself on first run.
- Your phone. The remote itself is one web page - nothing to install from an app store.
- The RemoVia bridge folder. Download the ZIP and keep the folder it makes (it is called removia-main).
- A free Tailscale account. Tailscale is the private, secure path between your phone and the Mac, so the remote works from anywhere. Step 3 sets it up.
- Each TV's own remote, once. Pairing is approved on the TV screen, one time per TV.

TVs RemoVia works with: Roku - Samsung - Google TV and Android TV sets (Sony, TCL, Hisense, Philips, Xiaomi, NVIDIA Shield) - Amazon Fire TV. LG (webOS) sets are found and saved, but cannot be controlled yet.

Step 2 - Install and start the bridge

1. Download the RemoVia bridge ZIP and unzip it. You get a folder named removia-main in your Downloads.
2. Open Terminal on the Mac.
3. Go to the folder:

```
cd ~/Downloads/removia-main
```

4. Start the bridge:

```
./start.sh
```

The first run builds its own Python environment - it takes a few minutes. Later runs start at once.

5. Keep that Terminal window open. The bridge runs while the window stays open.
6. On the first start only, the bridge prints your bridge access token. Save it somewhere safe. You type it into the remote page once per device. If you lose it, it is stored on the Mac in ~/.tv-remote/config.json.
7. macOS may ask whether to allow Python to accept incoming connections. Click Allow.

Protect the token.
The access token is a key to your TVs. Never share the token, and never share the ~/.tv-remote/config.json file - it holds the token. Anyone who has the token and a path to your bridge can control your TVs.

Step 3 - Connect the page securely (Tailscale)

The hosted page at removia.iecincglobal.com is a secure (https) page. Browsers block a secure page from calling a plain http:// address, so the page cannot reach the bridge on your home network directly. Tailscale fixes this: it gives the bridge a secure https address that only your own devices can reach.

1. Install Tailscale on the Mac and on your phone. Sign in with the same account on both - that is what links them.
2. Open the Tailscale admin console and enable HTTPS: go to console.tailscale.com/admin/dns, find HTTPS Certificates, and click Enable HTTPS.
3. On the Mac, in Terminal, run:

```
tailscale serve --bg 8765
```

4. The command prints your bridge's secure address. It looks like https://your-mac.tail1234.ts.net. Copy it.
5. On your phone, open removia.iecincglobal.com and tap the gear (Settings).
6. Set Bridge address to the ts.net address from step 4. Set Token to your bridge access token. Tap Save, then Test.
7. Success is a green "Connected" dot at the top of the page. Your remote now works from anywhere - home or away.

Nothing is exposed to the public internet.
Tailscale is a private network made only of your signed-in devices. No router changes and no port-forwarding are needed - and you should never set up port-forwarding for the bridge. At home, the bridge also serves the same remote page itself at http://<your-mac-ip>:8765 - a handy fallback.

Step 4 - Add your TV, and find its IP address

1. In RemoVia, tap + TV.
2. Tap Scan the network. When your TV appears, tap Add and give it a room name (for example, BedRoom).
3. If the scan misses a TV, add it by hand: pick the brand, type the TV's IP address, name it, and tap Add this TV.

How to find a TV's IP address (use the TV's own remote):

- Samsung: Settings > General > Network > Network Status > IP Settings. Read the IP Address line.
- Roku: Settings > Network > About. Read the IP address.

Enter the TV's own IP - never the Gateway.
The Samsung IP Settings screen shows both lines side by side: IP Address is the TV; Gateway is your router. On most home networks the gateway is 10.0.0.1. If you enter the gateway address, the bridge knocks on the router's door and pairing fails with ConnectionRefusedError. Always use the IP Address line.

Step 5 - Pair by brand

Pairing is a one-time approval on the TV itself. After that, the TV remembers RemoVia.

Roku - no pairing, one setting

Roku needs no pairing. It works right away - as long as control by apps is allowed. With the TV's own remote: Settings > System > Advanced system settings > Control by mobile apps > Network access = Permissive. If this is left on Limited, every button you press answers with error (403).

Samsung - accept once on the TV

1. Turn the TV fully on and put it on a normal screen. Close all menus - the pairing prompt cannot appear while Settings is open.
2. In RemoVia, select the TV and tap Start pairing.
3. A prompt appears on the TV: "RemoVia - Your permission is required to connect your device to TV." Press Allow with the TV's own remote.
4. Tap Done on the phone. That is it - one time only.

If no prompt appears: on the TV, open Settings > General > External Device Manager > Device Connect Manager and set Access Notification to Always On. Also make sure network remote is on: Settings > General > Network > Expert Settings > Power with Mobile. Menu names vary by model.

Google TV / Android TV - type the PIN

For Sony, TCL, Hisense, Philips, Xiaomi and NVIDIA Shield sets: select the TV and tap Start pairing. The TV shows a 6-digit PIN. Type the PIN in RemoVia and tap Pair with this PIN.

Fire TV - allow the ADB prompt

1. First, on the TV: Settings > My Fire TV > Developer Options > ADB Debugging = On.
2. In RemoVia, select the TV and tap Start pairing.
3. Allow the prompt on the TV. Tick "Always allow" if it is shown.
4. Tap Done on the phone.

LG (webOS)

RemoVia finds and saves LG sets, but cannot control them yet - there is no pairing step. Control for webOS is planned, not promised.

Step 6 - Troubleshooting

Error (403) on a Roku

Fix: On the TV: Settings > System > Advanced system settings > Control by mobile apps > Network access = Permissive.

ConnectionRefusedError

Fix: The bridge used the wrong IP (often the gateway, 10.0.0.1), or the TV is off. Read the TV's real IP Address in its Network settings (Step 4), re-add the TV with that address, and make sure the TV is on.

"The TV did not accept in time, or it is off"

Fix: TV fully on, all menus closed. Tap Start pairing again and press Allow on the TV promptly.

"Stopped answering"

Fix: Restart the bridge: in its Terminal window press Ctrl+C, then run ./start.sh again.

"Address already in use" when starting the bridge

Fix: The old bridge is still running. Stop it first: Ctrl+C in its Terminal window, or run: kill $(lsof -ti :8765) - then ./start.sh.

Red dot - the page cannot reach the bridge

Fix: Check Tailscale is on and signed in with the same account on phone and Mac. Check the bridge Terminal window is open. Check the bridge address in Settings matches the address tailscale serve printed.

Phone cannot connect at home

Fix: The macOS firewall may be blocking the bridge. System Settings > Network > Firewall > Options > allow incoming connections for Python.

Step 7 - Safety notes

- The bridge access token is a key to your TVs. Never post it, share it, or share the ~/.tv-remote/config.json file that holds it.
- Tailscale keeps everything private. Only devices signed in to your account can reach your bridge. Nothing in your house is exposed to the public internet, and no router port-forwarding is needed - never set up port-forwarding for the bridge.
- Pairing prompts appear on the TV itself. Press Allow only for a device you recognise. The prompt names the app - RemoVia.
- The bridge Mac is the door. Keep it on your home Wi-Fi and keep its Terminal window open while you use the remote.
- Pairing happens once per TV. To revoke a device later, remove it in the TV's own device settings.

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